DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | November 5, 2026 · *drifting stream*

LECTURE III

Data Engineering Fundamentals

This lecture in one slide

III Data Engineering Fundamentals

Data Sources
Data Formats
Data Models
Data Storage Engines and Processing
Modes of Dataflow
Batch Processing Versus Stream Processing

DATA
ENGINEERING
FUNDAMENTALS

Knowing how to handle a growing amount of data

Data Sources
DATA ENGINEERING FUNDAMENTALS
"or knowing how to collect, process, store, retrieve and process an increasing amount of data."

Data Sources
Data Formats, Structures, & Models
DATA ENGINEERING FUNDAMENTALS
"or knowing how to collect, process, store, retrieve and process an increasing amount of data."

Data Sources
Data Formats, Structures, & Models
DATA ENGINEERING FUNDAMENTALS
Data Storage, Retrieval & Processing.
...or knowing how to collect, process, store, retrieve and process an increasing amount of data.

Further reading: Kleppmann 2017

1

Data Sources

DATA SOURCES
Characteristics
Purposes
Processing Methods

DATA SOURCES

- Characteristics
- Purposes
- Processing Methods

... knowing your data lets you use your data more efficiently.

Users Input Data

Malformatted, any format, requires fast processing.

Users Input Data

Halfformatted, any format, requires post processing.

System Generated Data

Logs, System outputs, model predictions.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

Users Input Data
Halfformatted, any format, requires fast processing.

System Generated Data
Logs, System outputs, model predictions.

Log everything you can, store, are allowed to

Common practice to log everything you can.

... and you can store.

... and are allowed to.

User Input Data

Half-formatted, any format, requires fast processing.

System Generated Data

Logs, System outputs, model predictions.

Internal Databases

Enterprise applications and assets.

Common practice to log everything you can.
... and you can store.
... and are allowed to.

User Input Data
Halfformatted, any format, requires fast processing.

System Generated Data
Logs, System outputs, model predictions.

Internal Databases
Enterprise applications and assets.

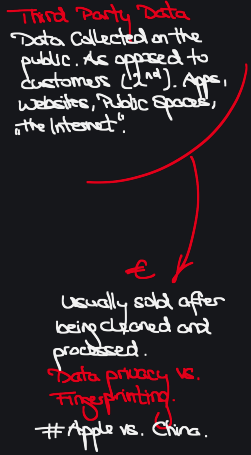
Third Party Data
Data collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces", "the Internet".

Third Party Data

"Data collected on the public. As opposed to customers (1st). Apps, websites, "Public Spaces, "the Internet".

€

Usually sold after being cleaned and processed.


Third Party Data
"Data Collected on the public. As opposed to customers (1st). Apps, Websites, "Public Spaces", "the Internet".
€
Usually sold after being cleaned and processed.
"Data privacy vs. Fingerprinting.
#Apple vs. China.

2

Data Formats

3

Data Models

4

Data Storage Engines and Processing

5

Modes of Dataflow

6

Batch Processing Versus Stream Processing

References

1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications.* Sebastopol, CA: O'Reilly Media, 2022.

2 Martin Kleppmann. *Designing Data-Intensive Applications.* Sebastopol, CA: O'Reilly Media, 2017.